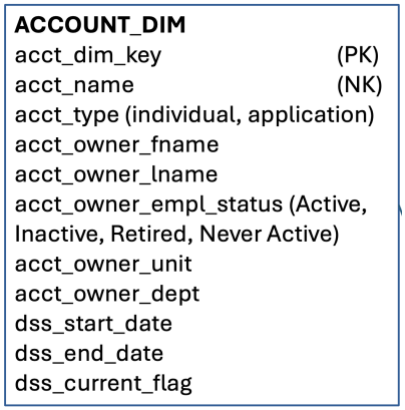
ACCOUNT_DIM recipe

For my data model, I'm creating an account_dim to track the "who". Who has access to what? Who is accessing what? The who is a database account, (individual, or application) owned by someone at Cornell. Note there are a two derived fields account_type & account_owner_empl_status. This is also a type 2 slowly changing dimension table, so the sponsors can easily identify when an account owner changes their unit or department.



Source-to-Target mapping:

Source table	Source column	Target table	Target column	Transformation / notes
Oracle_users	username	Account_dim	Acct_name	Group by username
Oracle_users	username	Account_dim	Acct_type	REGEXP_LIKE(username, '^.{2,}[0-9]+$') = 'Individual' ELSE 'Application'
Cu_person	First_name	Account_dim	Acct_owner_fname	Owner info for application accounts join cu_person through appid_owners.
Cu_person	Last_name	Account_dim	Acct_owner_lname	
Cu_person	active	Account_dim	Acct_owner_emp_status	A=Active, I=Inactive, R=Retired, N=Never Active
Cu_person	Unit_name	Account_dim	Acct_owner_unit	
Cu_person	Department_name	Account_dim	Acct_owner_dept	

Load Tables:

If you look at the source-to-target mappings, you will see there are two tables in the source_table column; oracle_users & cu_person. In the notes / transformations there is a comment about needing the appid_owners table to join oracle_users to cu_person, for application accounts. Create three load tables: load_oracle_users, load_cu_person, & load_appid_owners. Remember you will use your audit_src connection as the source for your load tables.

Stage Tables:

I am using multiple stage tables to deduplicate, join to source tables, and clean the data for my account_dim table.

STAGE_ACCOUNT_1: The only table containing all of the database accounts is oracle_users. If you query this table, you will notice there are many "usernames" which exist in multiple databases. For my account_dim table, I want the account name to be unique. So, I am creating a stage table containing a unique list of database accounts, this will be stage_account_1.

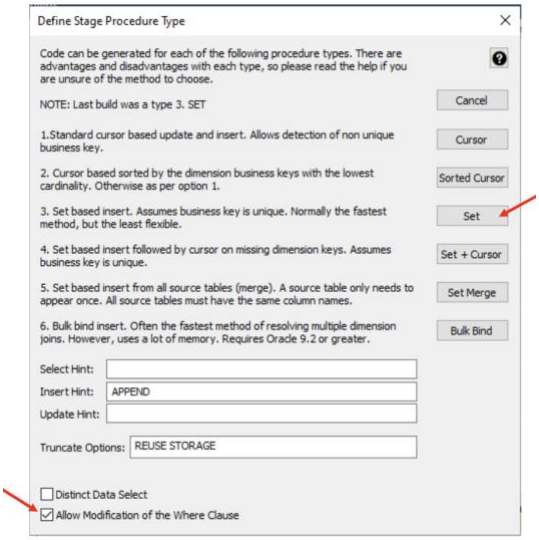
1) In left pane, double click "Stage Table", so Stage Table becomes the focus of the center pane.
2) In right pane, change connection from audit_src to warehouse. Then expand the load_oracle_users table by double clicking on it. Then pull the username column to the middle pane.
3) When prompted for the name of the stage table. I changed the name from the default of "stage_oracle_users" to "stage_account_1".
4) My stage_account_1 table definition looks like this:

stage_account_1 Columns

Column Name	Display Name	Data Type	Source Table	Source Column	Format	Nulls	Nu
username	username	varchar2(40)	load_oracle_users	username		Y	N
dss_create_time	dss create time	date		dss_create_time		Y	N
dss_update_time	dss update time	date		dss_update_time		Y	N

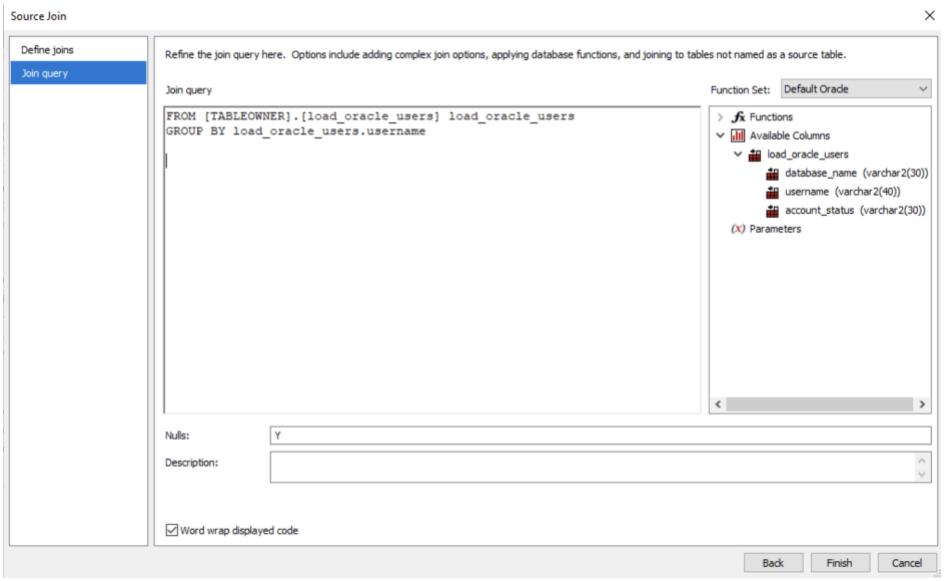
5) In the left pane, right click on stage_account_1 and select "create(recreate)".
6) In the left pane, right click on stage_account_1 and select "code > rebuild update procedure".

7) In the “Define Stage Procedure Type” screen, check the “Allow Modification of the Where Clause”, then select “Set”.



8) For the “Select Parameters” screen, simply click “OK”.
9) In the “Source Join” screen. Under the FROM clause, add the following “GROUP BY”.

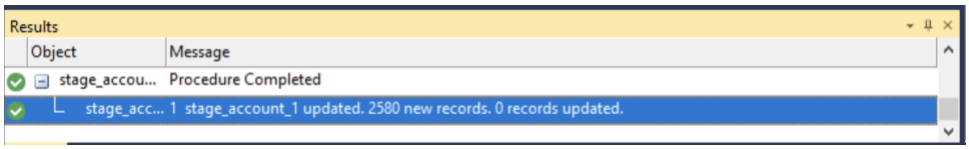
GROUP BY load_oracle_users.username



Then click “Finish”.

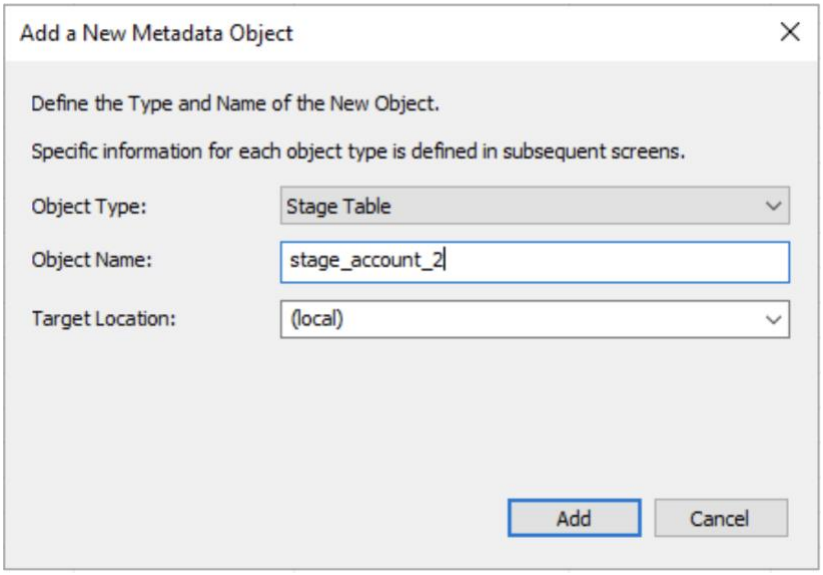
10) In left pane, right click on stage_account_1, then select “execute update procedure” to run your code.

Successful execution will look like:

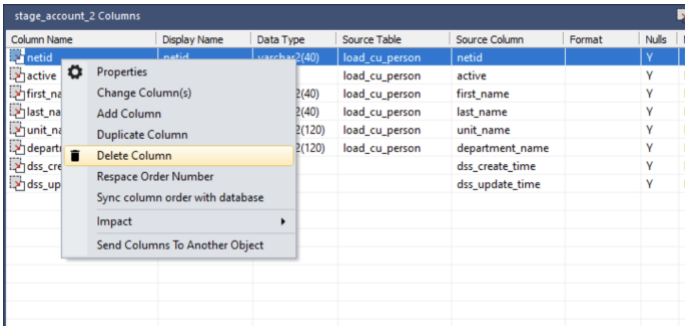


STAGE_ACCOUNT_2: Will contain only application IDs with owners defined in the load_appid_owners table.

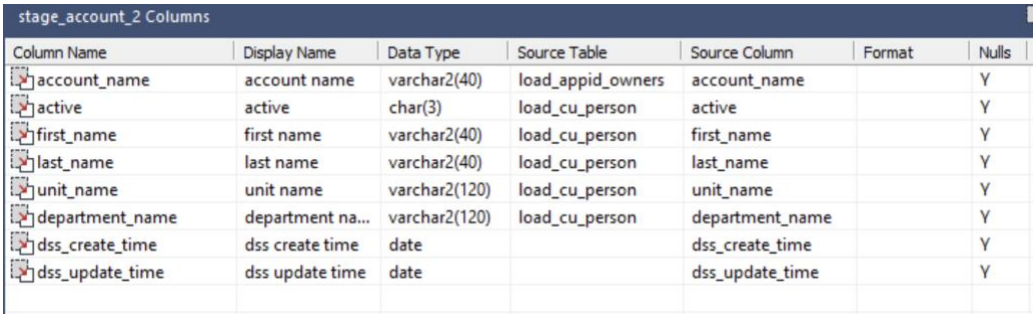
1) In left pane, double click “Stage Table”, so Stage Table becomes focus of middle pane.
2) In right pane, drag the load_cu_person table into middle pane.
3) In the “Add a New Metadata Object” screen, name object “stage_account_2, then click Add



4) In Stage Table properties screen, simply click OK.
5) Right click on column netid and select “Delete Column”. We will replace this with the application account name from the load_appid_owners table.



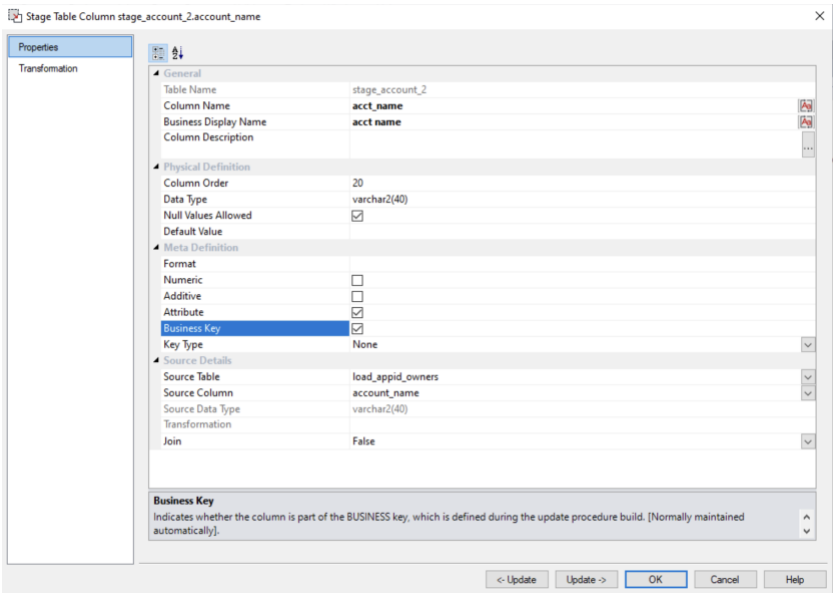
6) In right pane, double click your load_appid_owners table to expand it. Then drag the account_name column to the top of the center pane.
7) Should look like this, after dragging the account_name column to middle pane.



stage_account_2 Columns

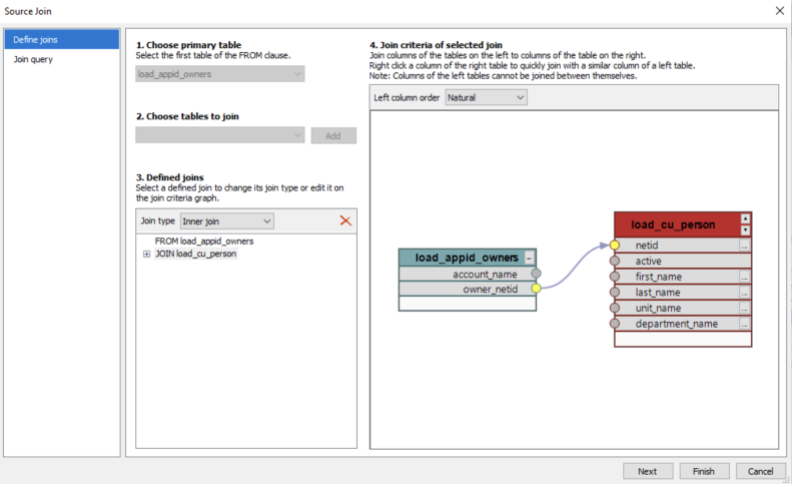
Column Name	Display Name	Data Type	Source Table	Source Column	Format	Nulls
account_name	account name	varchar2(40)	load_appid_owners	account_name		Y
active	active	char(3)	load_cu_person	active		Y
first_name	first name	varchar2(40)	load_cu_person	first_name		Y
last_name	last name	varchar2(40)	load_cu_person	last_name		Y
unit_name	unit name	varchar2(120)	load_cu_person	unit_name		Y
department_name	department na...	varchar2(120)	load_cu_person	department_name		Y
dss_create_time	dss create time	date		dss_create_time		Y
dss_update_time	dss update time	date		dss_update_time		Y

8) In middle pane, double click your account_name column, then rename to acct_name. Check the “Business Key” check box, then select OK.

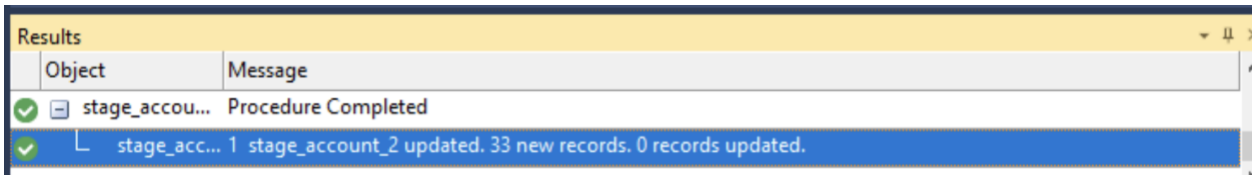


9) In left pane, right click “stage_account_2” table then select “create(recreate)” to build table in your database account.
10) In left pane, right click “stage_account_2” table then select “code>rebuild update_stage_account_2”
11) Select “Set”
12) For “Select Parameters” screen, simply click OK
13) In the “Source Join” screen, click Add, then drag join line from owner_netid to netid. Note that this join should be an “inner join”. Then click “Finish”.

ACCOUNT_DIM recipe

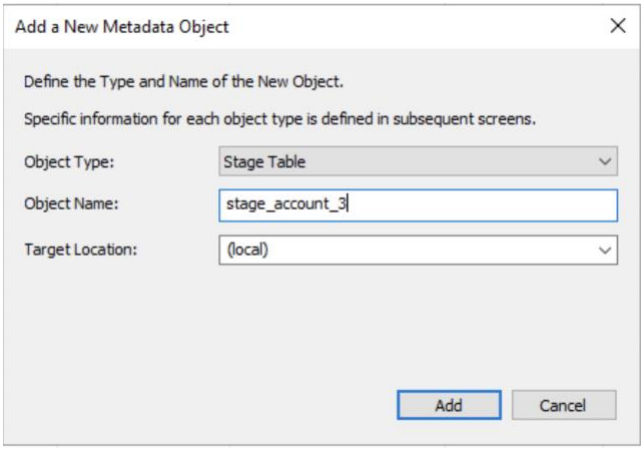


14) In left pane, right click “stage_account_2, then select “execute update procedure”.
15) Result should look like:

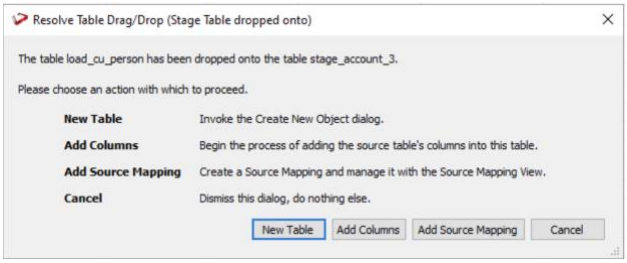


STAGE_ACCOUNT_3: This stage table will contain ALL database account from load_users (with the exception of the 33 application account with defined owners). We will join stage_account_1 to load_cu_person (left outer join) to get the account owner data where available.

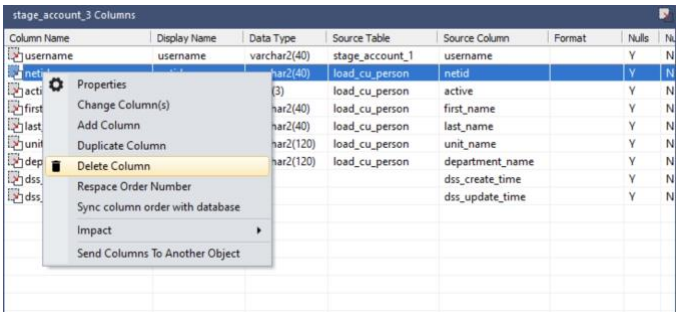
1) In left pane, double click “Stage Table”, so Stage Table becomes focus of middle pane.
2) In right pane, click the “blue Sunglasses” icon to refresh your warehouse connection.
3) Drag your stage_account_1 table from the left pane to the center pane.
4) In the “Add a New Metadata Object” screen, name your new table “stage_account_3”, then click Add.



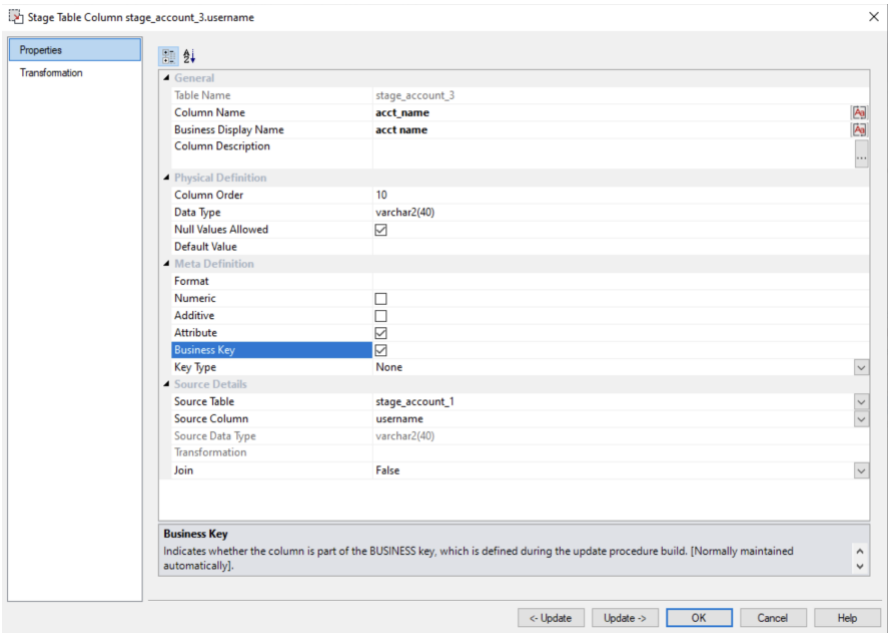
5) For the “Stage Table” screen, simply click “OK”.
6) In right pane, select your “load_cu_person” table and drag to the middle pane.
7) If you see a screen “Resolve TableDrag/Drop”, click “Add Columns”.



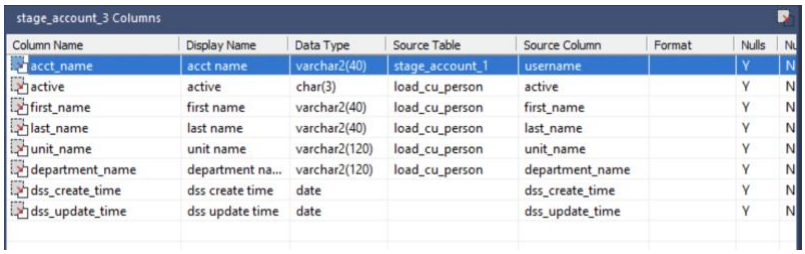
8) In middle pane, right click the netid column and select “Delete Column”.



9) If prompted to delete 1 column, click “yes”.
10) In middle pane, double click the “username” column.
11) Rename the column name to “acct_name” and the business display name to “acct name”. Also, check the “Business Key” checkbox. Then click OK.

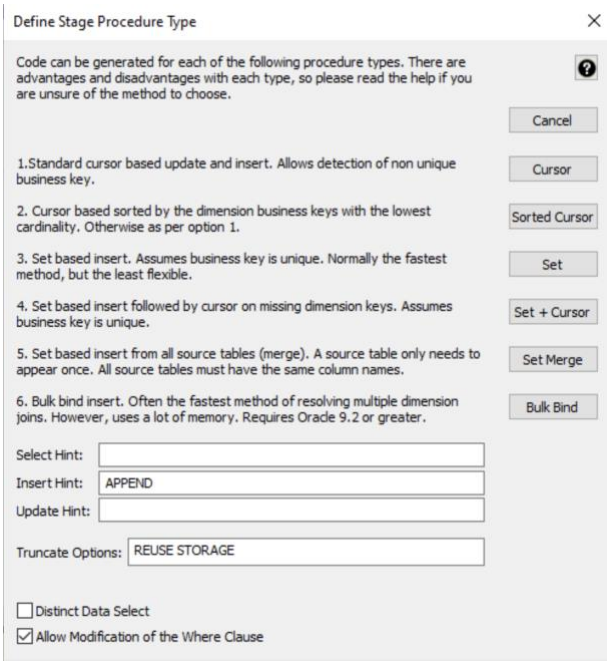


12) Your middle pane (table columns) should look like this:



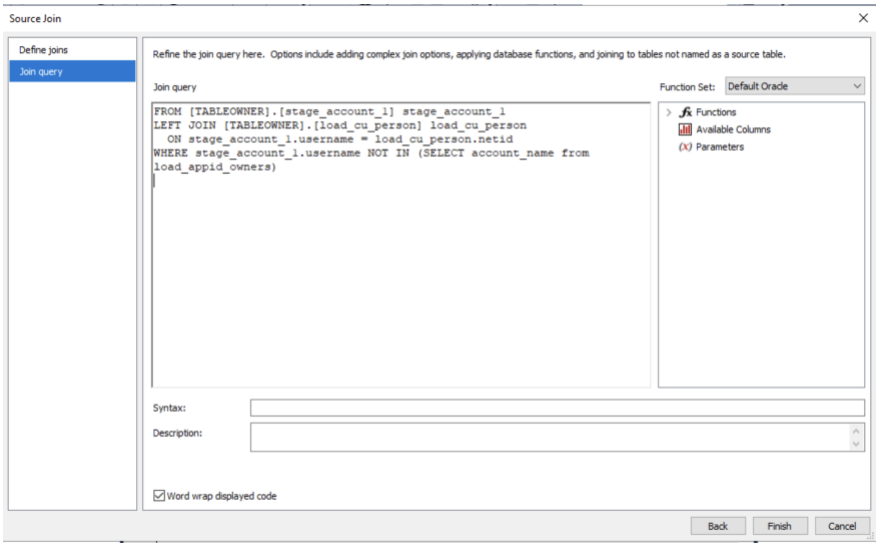
Column Name	Display Name	Data Type	Source Table	Source Column	Format	Nulls	Nu
acct_name	acct name	varchar2(40)	stage_account_1	username		Y	N
active	active	char(3)	load_cu_person	active		Y	N
first_name	first name	varchar2(40)	load_cu_person	first_name		Y	N
last_name	last name	varchar2(40)	load_cu_person	last_name		Y	N
unit_name	unit name	varchar2(120)	load_cu_person	unit_name		Y	N
department_name	department na...	varchar2(120)	load_cu_person	department_name		Y	N
dss_create_time	dss create time	date		dss_create_time		Y	N
dss_update_time	dss update time	date		dss_update_time		Y	N

13) In the left pane, right click on stage_account_3, then select “create(recreate)” to build your new stage_account_3 table.
14) In left pane, right click stage_account_3 table, then select “code> rebuild update procedure”.
15) On the “Define Stage Procedure Type” screen, check the “allow modification of the Where Clause” checkbox, then select “Set”.



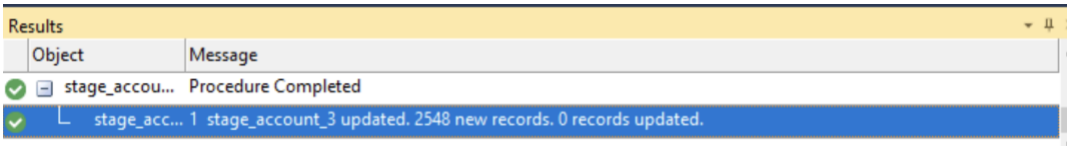
16) For the “Select Parameters” screen, simply select OK.
17) For the “Source Join” screen, click Add, then drag the join line from username to netid. Change the Join Type to “Left outer Join”. Then click Next.
18) In the “Join Query” screen add the following code below the join code. This code will exclude the account names from the account_owners table.

WHERE stage_account_1.username NOT IN (SELECT account_name from load_appid_owners)



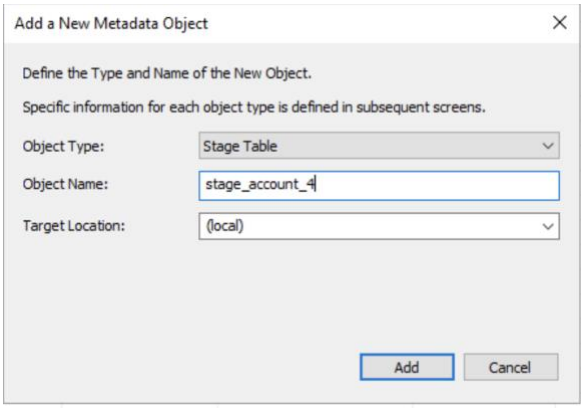
Then click “Finish”.

19) In left pane, right click stage_account_3, then select “execute update procedure”. Results should look like this:

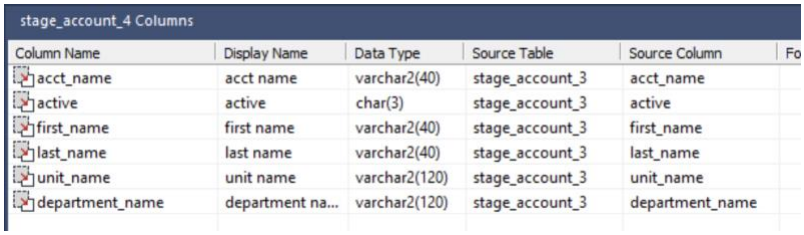


STAGE_ACCOUNT_4: This stage table will contain the result of merging stage_account_2 & stage_account_3 via a UNION.

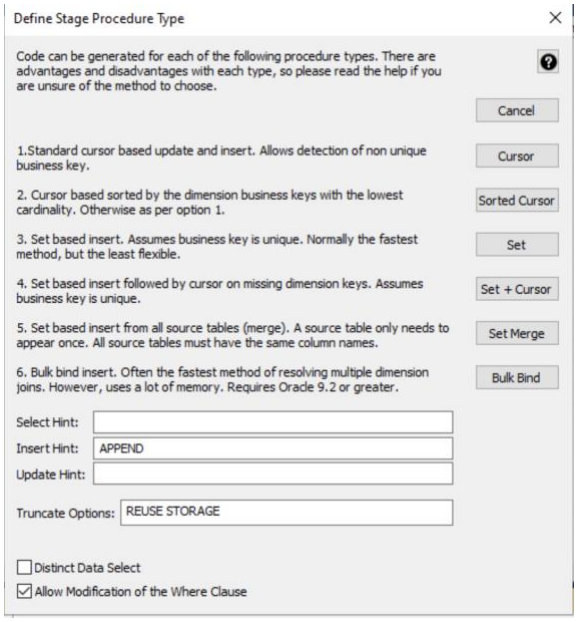
1) In left pane, double click “Stage Table”, so Stage Table becomes the focus of the middle pane.
2) In right pane, click on the blue sunglasses, so your warehouse connection refreshes. You need to refresh your warehouse connection, so you can see your new stage tables.
3) In right pane, select your stage_account_3 table and drag to the middle pane.
4) In the “Add a New Metadata Object” screen, rename Object name to “stage_account_4”, then select Add.



5) For the “Stage Table” screen, simply click OK.
6) In middle pane, right click column dss_create_time and select “delete column”.
7) In middle pane, right click column dss_update_time and select “delete column”.
8) Your middle pane should look like this:



9) In left pane, right click “stage_account_4”, then select “create(recreate)”.
10) In left pane, right click “stage_account_4” then select “code>rebuild update procedure”.
11) For “Define Stage Procedure Type” screen, click the “Allow Modification of the WhereClause” checkbox, then select Set.

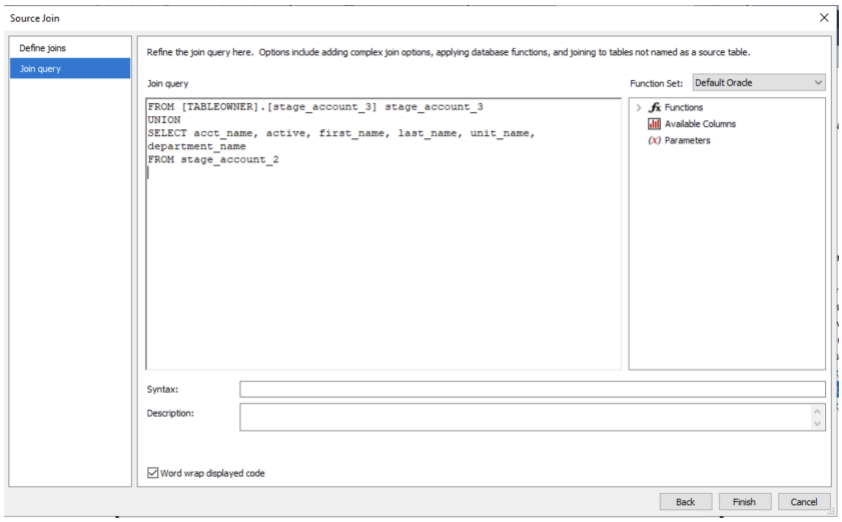


12) For the “Select Parameters” screen, simply select “OK”.
13) In the “Source Join” code screen, add the following code. This will pull in the data from the stage_account_2 table.

```
UNION

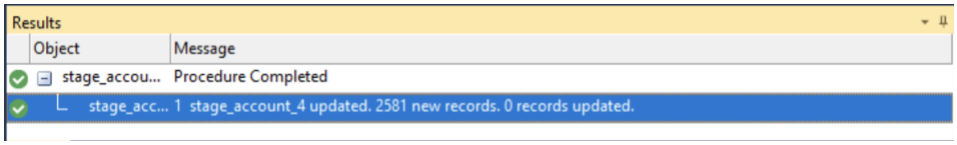
SELECT acct_name, active, first_name, last_name, unit_name, department_name

FROM stage_account_2
```



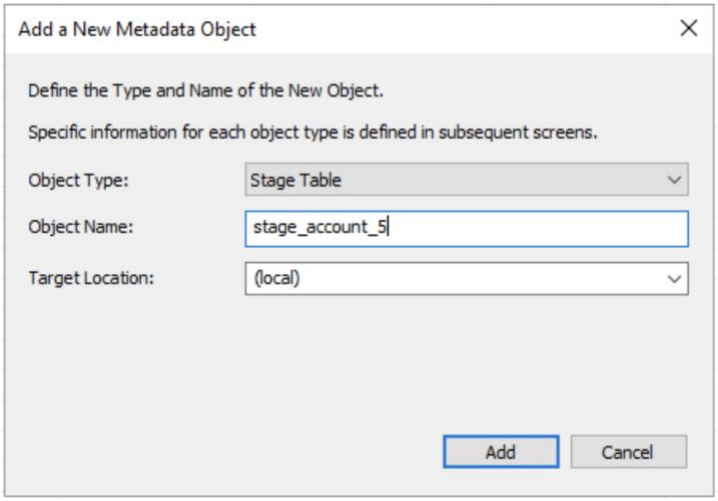
Then click “Finish”.

14) In left pane, right click stage_account_4 table, then select “execute update procedure”. Should look like:



STAGE_ACCOUNT_5: This stage table will be used to add any derived attributes, transformations, and rename columns to what will be shown in dimension table.

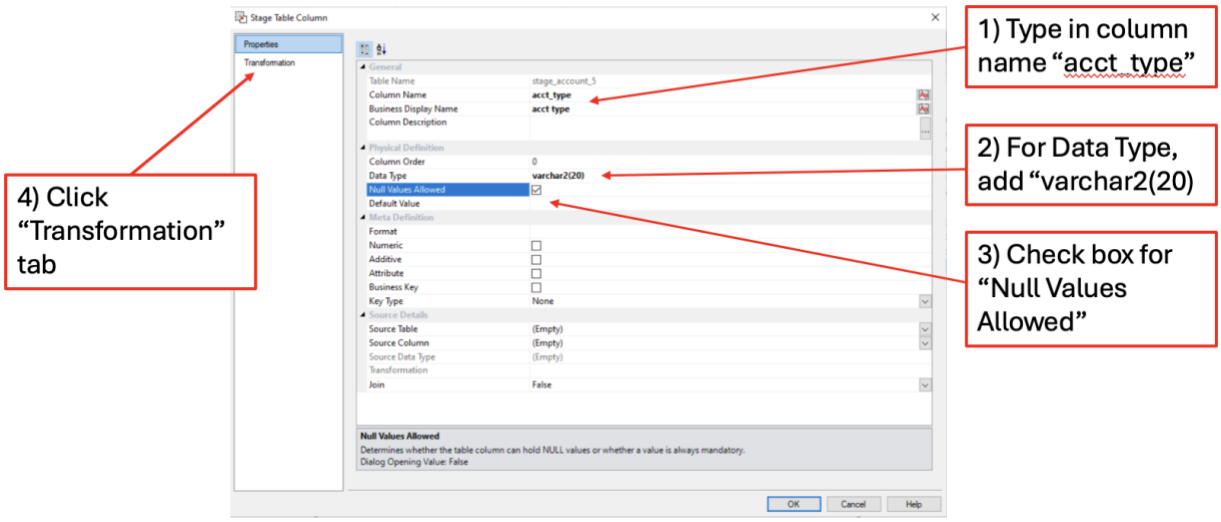
1) In left pane, double click “Stage Table”, so Stage Table becomes the focus of the middle pane.
2) In the right pane, click the blue sunglasses to refresh your data warehouse connection. So, you can see stage_account_4 table.
3) In the right pane, select stage_account_4 table and drag to middle pane.
4) For “Add a New Metadata Object” screen, rename object_name to “Stage_account_5”, then click Add.



5) For the “Stage Table” screen, simply click OK.
6) Here’s the source-to-target mapping table, from page 1, for the ACCOUNT_DIM table we are building. We need to name all of the columns the same in STAGE_ACCOUNT_5, and add any needed transformations.

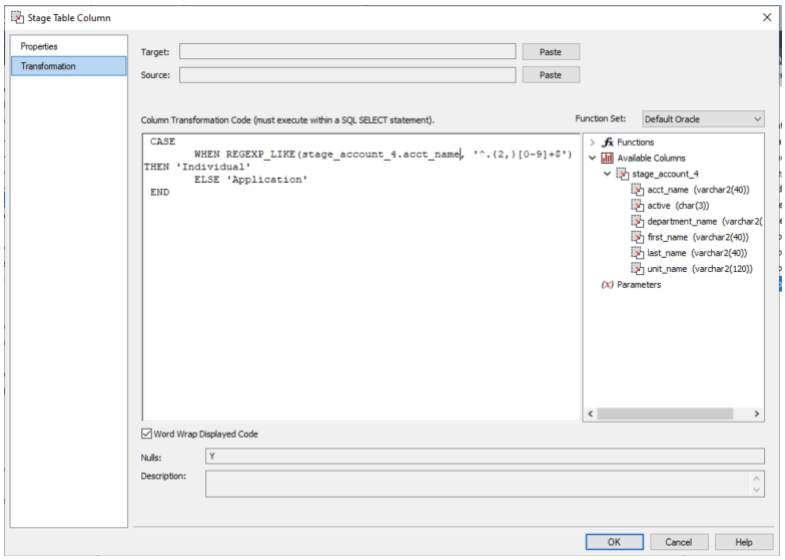
Source table	Source column	Target table	Target column	Transformation / notes
Oracle_users	username	Account_dim	Acct_name	Group by username
Oracle_users	username	Account_dim	Acct_type	REGEXP_LIKE(username, '^.{2,}[0-9]+$') = ‘Individual’ ELSE ‘Application’
Cu_person	First_name	Account_dim	Acct_owner_fname	Owner info for application accounts join cu_person through appid_owners.
Cu_person	Last_name	Account_dim	Acct_owner_lname	
Cu_person	active	Account_dim	Acct_owner_emp_status	A=Active, I=Inactive, R=Retired, N=Never Active
Cu_person	Unit_name	Account_dim	Acct_owner_unit	
Cu_person	Department_name	Account_dim	Acct_owner_department	

7) Right click anywhere in the middle pane, then select “Add Column”.
8) In the “Stage Table Column” screen, type in the column name acct_type. For Data Type add “varchar2(20). Also, check the “Null Values Allowed” checkbox. Finally, click on the “transformation” tab. *See diagram below.*



9) In the Transformation code window past the following code. This will set any account name with a netid format to ‘Individual’ and all others to ‘Application’.

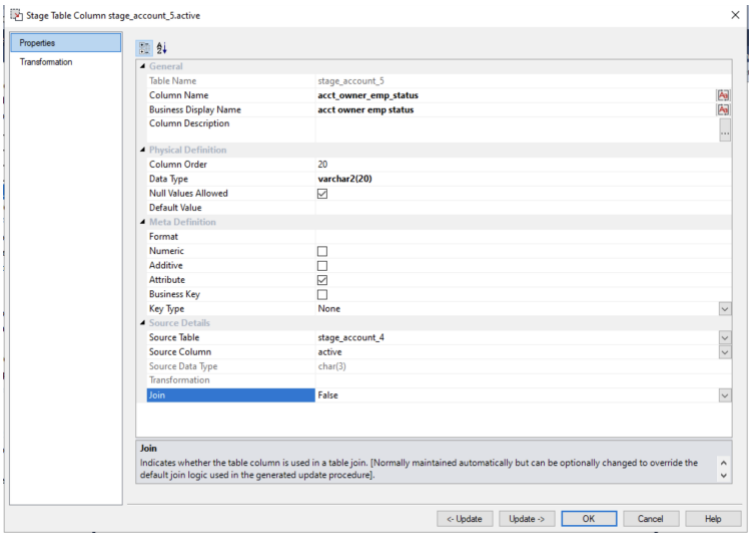
```
CASE
    WHEN REGEXP_LIKE(acct_name, '^.{2,}[0-9]+$') THEN 'Individual'
    ELSE 'Application'
END
```



Click OK, once code is in place.

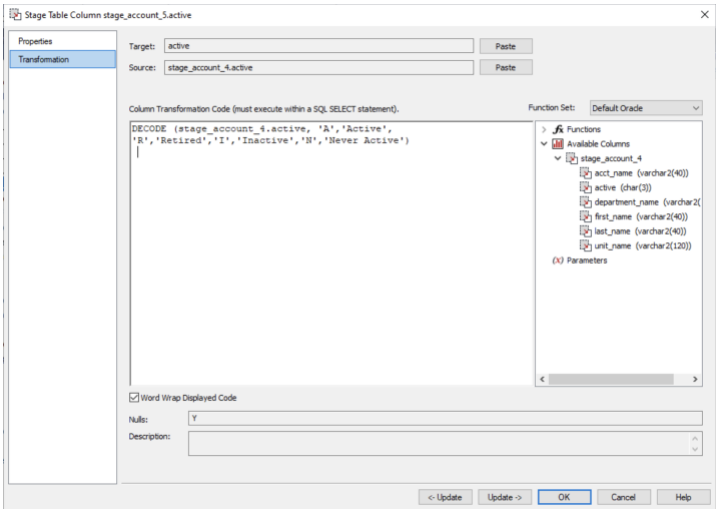
10) In middle pane, double click on the “active” column.

11) Rename “active” column to “acct_owner_emp_status”, then change the data type to “varchar2(20)”. The click the “Transformation” tab on the upper left corner.



12) In the transformation code window, past the following code. This will transform the codes into meaningful descriptive words.

DECODE (RTRIM(stage_account_4.active), 'A','Active', 'R','Retired','I','Inactive','N','Never Active')



Once code is in place, click OK.

13) Now rename the remaining columns to match Your data model. For my data model, renamed the following columns:
First_name to acct_owner_fname

Last_name to acct_owner_lname
Unit_name to acct_owner_unit
Department_name to acct_owner_department

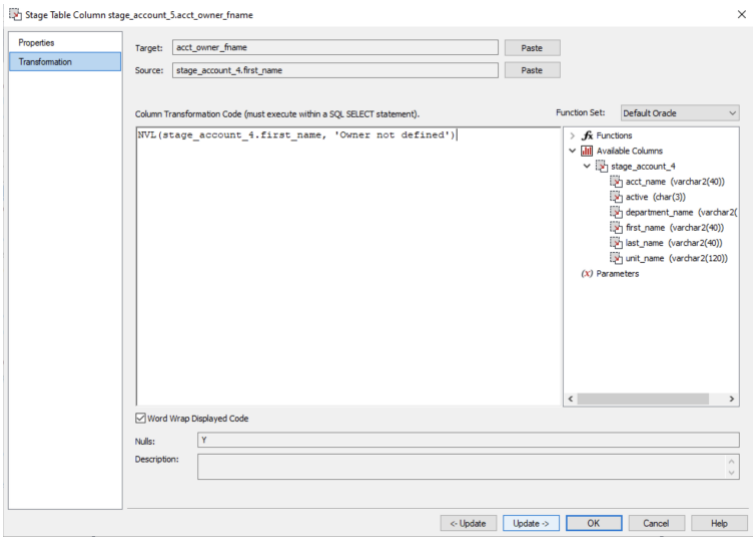
14) I then re-ordered the columns to match my data model. For my data model the final result looks like.

stage_account_5 Columns

Column Name	Display Name	Data Type	Source Table	Source Column	Format	Nulls	Nu.
acct_name	acct name	varchar2(40)	stage_account_4	acct_name		Y	N
acct_type	acct type	varchar2(20)				Y	N
acct_owner_fname	acct owner fna...	varchar2(40)	stage_account_4	first_name		Y	N
acct_owner_lname	acct owner lna...	varchar2(40)	stage_account_4	last_name		Y	N
acct_owner_emp_status	acct owner em...	varchar2(20)	stage_account_4	active		Y	N
acct_owner_unit	unit name	varchar2(120)	stage_account_4	unit_name		Y	N
acct_owner_department	acct owner de...	varchar2(120)	stage_account_4	department_name		Y	N
dss_create_time	dss create time	date		dss_create_time		Y	N
dss_update_time	dss update time	date		dss_update_time		Y	N

A few notes: the dss_ columns are delivered with WhereScape for auditing the records. You may drop them if you like, since this is a proof of concept only. Also, the surrogate key (PK) will be added by WhereScape when we build the dimension table.

15) Now we need to create a transformation for the account owner columns, so there is a value of “Owner not defined”, if there is a null value. Unfortunately, we don’t have owners mapped to all of the application accounts yet, so we want something other than null.
16) Double click the acct_owner_fname column, then in the “Stage Table Column” screen, click the “Transformation” tab in the upper left corner.
17) In the Transformation code window past the following code:
NVL(stage_account_4.first_name, 'Owner not defined')



Once code is in place, click OK.

18) Repeat for the acct_owner_lname column. Here's the code for the transformation:

NVL(stage_account_4.last_name,'Owner not defined')

19) Repeat for the acct_owner_emp_status column. Here's the code for the transformation:

NVL(stage_account_4.unit_name, 'Owner not defined')

NVL(DECODE (RTRIM(stage_account_4.active), 'A','Active',
'R','Retired','I','Inactive','N','Never Active'), 'Owner not defined')

20) Repeat for the acct_owner_unit column. Here's the code for the transformation:
NVL(stage_account_4.unit_name, 'Owner not defined')

21) Repeat for the acct_owner_department column. Here's the code for the transformation:

NVL(stage_account_4.department_name, 'Owner not defined')

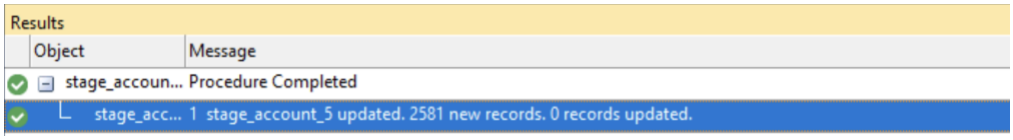
22)In left pane, right click the stage_account_5 table, then select "create(recreate)".

23) In left pane, right click the stage_account_5 table, then select "code > rebuild update procedure"

24) In the "Define Stage Procedure Type" screen, click Set.

25) In the "Select Parameters" screen, simply select OK.

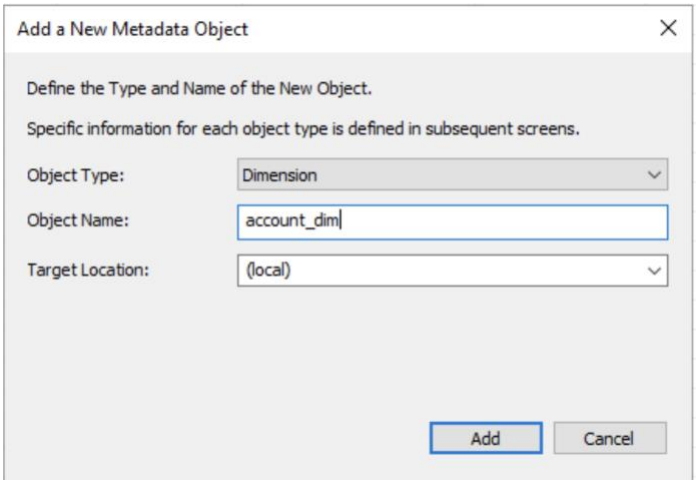
26) In the left pane, right click stage_account_5, then select "execute update procedure". Results should look like:



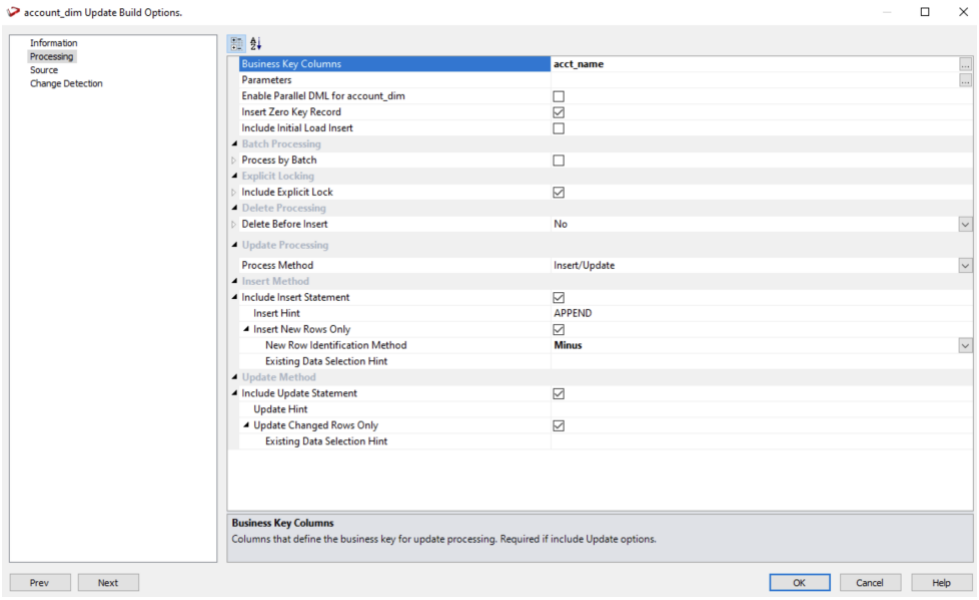
27)In left pane, right click stage_account_5, then select “display data”. Do you see data in all of the columns, for all rows? Are the values as expected? Do you see any records with “Owner not defined”? Hopefully, this means your transformations are working.

ACCOUNT_DIM:

1) In the left pane, double click dimension, so dimension is the focus of the middle pane.
2) In right pane, click on the blue sunglasses, to refresh your data warehouse connection.
3) In right pane, select the stage_account_5 table and drag into the middle pane.
4) In the ‘add a Metadata Object” screen, rename the “object Name” to account_dim, then click OK.



5) In the “Dimension Type” screen, select “Slowly Changing” this is for the Type 2 dimension, to meet the requirement regarding detecting when an account owner changes their status, unit, or department.
6) For the “Dimension account_dim” screen, simply click OK.
7) For the “Create Database Table” screen, select “Create”.
8) In the left pane, right click “account_dim”, then select “code> rebuild update procedure”
9) In the “Procedure Build Type” screen, select “OK”
10) In the “account_dim Update Key Columns” screen, add “acct_name” as the business key. Then click OK.

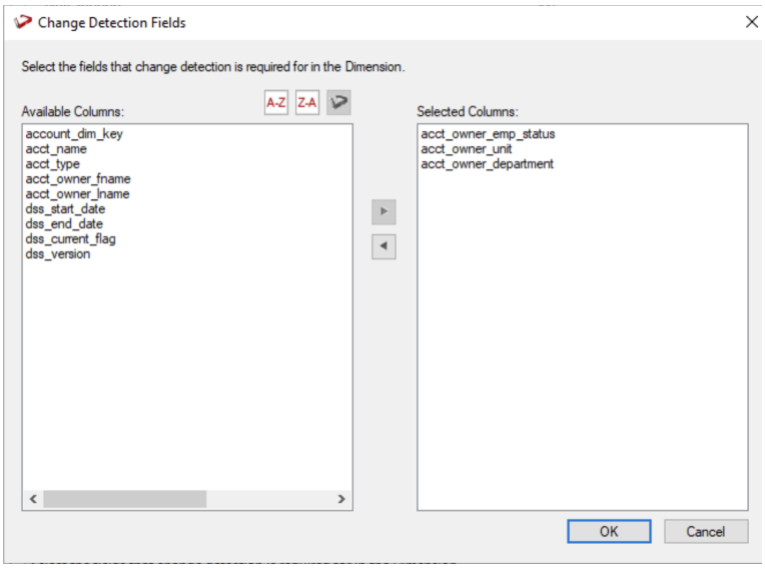


11) After click OK, you will see the following message:



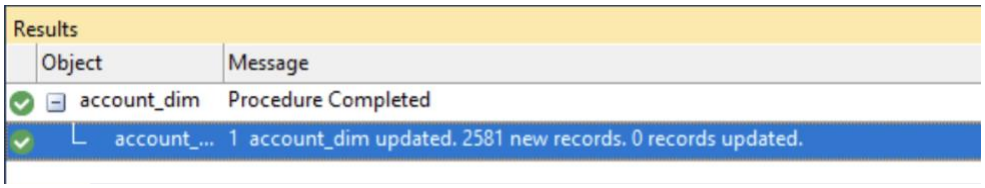
Simply click OK.

12) In the “Change Detection Fields” screen, select the following columns, from the left column and move to the right column “selected columns”. These are the columns which will be tracked in the type 2 Slowly Changing Dimension.
    a. Acct_owner_emp_status
    b. Acct_owner_unit
    c. Acct_owner_department



Click OK.

13) Back in the "account_dim Update Build Options" screen, click OK.
14) In left pane, right click account_dim, then select "execute update procedure".



15) In left pane, right click account_dim, then select "display data". How does the data look? You may want to run some test queries against your account_dim table via SQL Developer, but you should be good now.

ACCOUNT_DIM recipe

account_dim Columns

Column Name	Display Name	Data Type	Source Table	Source Column
account_dim_key	account dim key	integer		account_dim_key
acct_name	acct name	varchar2(40)	stage_account_5	acct_name
acct_type	acct type	varchar2(20)	stage_account_5	acct_type
acct_owner_fname	acct owner fna...	varchar2(40)	stage_account_5	acct_owner_fname
acct_owner_lname	acct owner lna...	varchar2(40)	stage_account_5	acct_owner_lname
acct_owner_emp_status	acct owner em...	varchar2(20)	stage_account_5	acct_owner_emp_s...
acct_owner_unit	unit name	varchar2(120)	stage_account_5	acct_owner_unit
acct_owner_department	acct owner de...	varchar2(120)	stage_account_5	acct_owner_depart...
dss_start_date	dss start date	date		dss_start_date
dss_end_date	dss end date	date		dss_end_date
dss_current_flag	dss current flag	char(1)		dss_current_flag
dss_version	dss version	integer		dss_version
dss_create_time	dss create time	date		dss_create_time
dss_update_time	dss update time	date		dss_update_time

